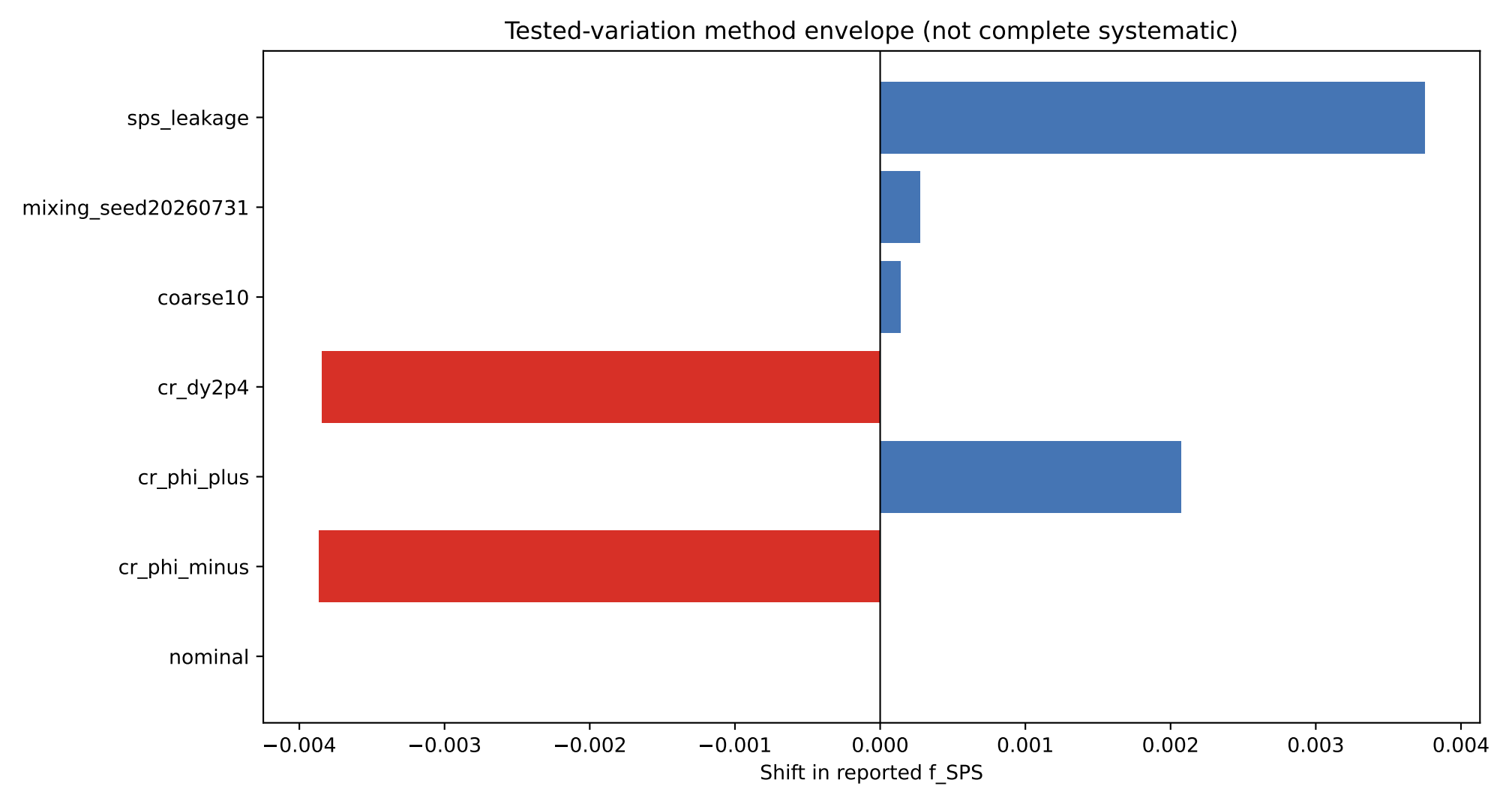
# H009: statistical propagation and closure

| test | result | scope |
|---|---|---|
| E1 joint stat | f_SPS = 0.807527 +/- 0.018829 | 50,000 toys |
| E1 Data-only SD | 0.018435 | reported estimator |
| E1 mix-only SD | 0.003794 | reported estimator |
| E2 truth 0.5 | bias +0.000945; pull width 0.919 | 30/30 complete |
| E2 truth 0.8 | bias +0.002891; pull width 1.191 | 30/30 complete |
| E2 truth 0.2 | stress test failed | 9/30; no coverage claim |


Tested-variation method envelope (not complete systematic)
sps_leakage
mixing_seed20260731
coarse10
cr_dy2p4
cr_phi_plus
cr_phi_minus
nominal
−0.004
−0.003
−0.002
−0.001
0.000
0.001
0.002
0.003
0.004
Shift in reported f_SPS

# E4 uncertainty contributions

| source | sigma_eff effect | treatment |
|---|---|---|
| f_DPS statistical | 9.78% | included_stat |
| f_DPS tested-method | 1.95% | included_syst_direction_swapped |
| CMS pair cross-section statistical | 1.61% | included_stat |
| CMS pair non-BR systematic | 6.68% | included_syst |
| CMS pair correlated BF removed | 0.00% | removed_BR_cancels |
| ATLAS single-Jpsi statistical | 0.19% | included_stat |
| ATLAS bin systematic | 1.69% | included_syst_uncorrelated_bins |
| ATLAS luminosity | 2.26% | included_syst_independent |
| BR cancellation | 0.00% | exact_in_visible_cross_section_formula |